

GW data analysis and low-latency notices

Compact Binary Coalescence (CBC)

Burst

	CBC	Burst
Source Model	Assumes specific waveforms from inspiral–merger–ringdown of compact binaries (BBH, BNS, NSBH)	No prior model — targets <i>unmodeled</i> or poorly modeled transients, e.g., core collapse SN, NS glitches...
Methodology	<i>Matched filtering</i> against a template bank of modeled waveforms	<i>Excess power / coherent reconstruction</i> in time–frequency domain
Main Pipelines	PyCBC, GstLAL, MBTA, SPIIR, AFrame, cWB BBH	cWB (Coherent WaveBurst), MLy
Public parameters	P_astro, EM_bright classification, CGMI, FAR, sky localization map	FAR, sky localization map

Some useful definitions

False Alarm Rate (FAR): defines how frequently a noise event is detected with the same observational properties of the GW candidate

- A. An event is **released publicly** if **FAR < 2 / day**
- B. An event is labeled as **SIGNIFICANT = True** if
 - **FAR < 1 / 7 months** if found by a **CBC** pipeline
 - **FAR < 1 / 3 years** if found by a **Burst** pipeline
- C. Otherwise is labeled as **SIGNIFICANT = False**. For non-significant events, no further checks on data quality or parameter estimation is performed, unless an external EM coincidence promotes the significance of the GW candidate (come to my next tutorial on RAVEN !!)

Signal to Noise Ratio (SNR): Identifies how loud is the signal with respect to the typical level of noise. Correlated with the significance of the event, but it is not a statistical measure of how we are confident about its astrophysical origin (glitches in the data can mimic high SNR GW signals, but a correct filtering of the pipelines should downrank them assigning high FAR)